Kalman Filter Signal Denoising

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***Abstract*—Noise is one of the common problems encountered in electronic and communication systems. It can distort a signal and make the original information difficult to recover. In this project, a Kalman filter is implemented in MATLAB to study how effectively it can estimate a signal affected by noise. A test signal containing three sinusoidal components is generated, and Gaussian noise is added to simulate noisy measurements. The filter estimates the signal by predicting its next state and correcting that prediction whenever a new measurement arrives. Its performance is studied using time-domain plots, frequency spectra, Mean Square Error (MSE), Root Mean Square Error (RMSE), and Signal-to-Noise Ratio (SNR). The results show that the chosen filter settings do not improve the signal quality in this experiment. The filtered output has a larger error than the noisy input, showing that the model and its parameters need further adjustment. The study helps explain how a Kalman filter works and why selecting a suitable model is important for obtaining useful results.**

***Index Terms*—Kalman filter, signal denoising, digital signal processing, state-space model, Gaussian noise, MSE, RMSE, SNR, MATLAB.**

I. Introduction

Signals play an important role in electronic systems, mobile communication, audio processing, and many other engineering applications. Ideally, a signal should reach the receiver without any unwanted disturbance. In real situations, however, this is rarely possible. Noise may enter a system through electronic components, sensors, transmission channels, or interference from the surrounding environment. Even when the original signal is clear, these disturbances can change its waveform and reduce the accuracy of the information being received.

For this reason, signal denoising is an important part of digital signal processing. The main challenge is to reduce the unwanted noise without removing the useful information present in the signal.

One method that can be used for this purpose is the Kalman filter. R. E. Kalman introduced this method in 1960 as a way to estimate the state of a system from uncertain measurements [1]. Since then, Kalman filtering has found applications in navigation, robotics, tracking, control systems, and signal processing.

The basic idea behind the Kalman filter is quite straightforward. Instead of relying entirely on a noisy measurement, the filter first predicts what the signal is expected to do. When the next measurement arrives, it compares that measurement with its prediction and adjusts the estimate accordingly. This process continues for every new sample. The Kalman gain determines how much the estimate should change in response to the measurement.

The accuracy of this method depends on how well the mathematical model represents the actual signal. If the model describes the signal reasonably well, the filter can combine its predictions with the measurements to obtain a useful estimate. If the model is unsuitable, however, the filter may produce an estimate that is less accurate than the original noisy input.

In this project, a Kalman filter is implemented in MATLAB and tested using a signal made up of three sinusoidal components. Gaussian noise is added with a target input SNR of 5 dB. A two-state model is used, where the two states represent the signal amplitude and its rate of change.

The original signal, noisy signal, and filtered output are then compared using plots and numerical performance measures. The main purpose is to understand the working of the Kalman filter, observe its behavior on a multi-frequency signal, and identify the reasons why the selected implementation may not give the expected improvement.

II. Problem Formulation

In this project, the main objective is to recover a signal that has been affected by random noise. The original signal is considered the reference, while the noisy signal represents the measurement available to the filter.

The measured signal is written as

$$y[k] = x[k] + v[k] \tag{1}$$

where $x[k]$ is the original signal, $y[k]$ is the noisy measurement, and $v[k]$ is the added Gaussian noise.

The task is to obtain an estimate of $x[k]$ from the available measurements. Ideally, the estimated signal should follow the original waveform closely while containing fewer unwanted variations.

To solve this problem using a Kalman filter, the signal is represented with a state-space model. The state equation describes how the signal state changes from one sample to the next, while the measurement equation explains how the state is related to the observed value.

The state-space equations are

$$X[k] = FX[k-1] + w[k] \tag{2}$$

and

$$Z[k] = HX[k] + v[k] \tag{3}$$

Here, $X[k]$ represents the state vector, F is the state transition matrix, and H is the measurement matrix. The term $w[k]$ represents process noise, while $v[k]$ represents measurement noise.

Since the original signal is available in the simulation, the filtered output can be compared directly with it. This makes it possible to calculate the estimation error and check whether the filter actually improves the signal.

An important part of this study is therefore to examine not only how the filter operates, but also how the choice of model and filter parameters affects its final performance.

III. Signal and Noise Model

Before applying the Kalman filter, a test signal must be generated and noise must be added to it. MATLAB is used for both tasks so that the original and noisy signals can be compared under controlled conditions.

A. Sampling Parameters

The sampling frequency is selected as

$$F_s = 1000 \text{ Hz}. \tag{4}$$

This means that the signal is represented by 1000 samples per second. The sampling period is

$$T = \frac{1}{F_s}. \tag{5}$$

A total of 5000 samples are used in the experiment. Therefore, the simulation covers approximately 5 seconds of signal data.

The test signal contains three sinusoidal components at 100 Hz, 250 Hz, and 400 Hz. Their amplitudes are 1.0, 0.5, and 0.3, respectively. The target input SNR is set to 5 dB.

The selected parameters are listed in Table I.

TABLE I
Simulation Parameters

Parameter	Value
Sampling Frequency	1000 Hz
Number of Samples	5000
Simulation Duration	5 s
Input SNR	5 dB
Frequency f_1	100 Hz
Frequency f_2	250 Hz
Frequency f_3	400 Hz
Amplitude A_1	1.0
Amplitude A_2	0.5
Amplitude A_3	0.3

B. Original Signal

The original signal is formed by adding three sine waves together. Each component has a different frequency and amplitude, giving the combined signal a more varied waveform.

The signal is represented as

$$x(t) = \sin(2\pi f_1 t) + 0.5\sin(2\pi f_2 t) + 0.3\sin(2\pi f_3 t) \tag{6}$$

where

$$f_1 = 100 \text{ Hz}, \qquad f_2 = 250 \text{ Hz}, \qquad f_3 = 400 \text{ Hz}. \tag{7}$$

The three components are combined to create a signal that can be used to test the filter under different rates of signal variation. The original waveform is stored so that the estimated signal can later be compared against it.

C. Gaussian Noise

In a practical system, noise may be introduced at different stages of signal generation, measurement, or transmission. To represent this effect in the simulation, zero-mean Gaussian noise is generated using MATLAB's `randn` function.

The required noise power is calculated from the desired input SNR:

$$P_{\text{noise}} = \frac{P_{\text{signal}}}{10^{\text{SNR}_{\text{input}}/10}}. \tag{8}$$

The noisy measurement is then obtained by adding the noise to the original signal:

$$y[k] = x[k] + v[k]. \tag{9}$$

The target input SNR is set to 5 dB. This creates a useful test case because the original waveform is still present, but the added noise makes direct observation less accurate.

IV. Kalman Filter Methodology

The Kalman filter works by repeatedly predicting the state of a system and correcting that prediction using the latest measurement. In this way, the estimate is updated as new data becomes available rather than being calculated only once at the end of the experiment [1], [2].

For the present study, a simple two-state model is used. Although this model is easy to implement, its suitability for a multi-frequency signal must be examined carefully.

A. State Vector

The state vector is defined as

$$X[k] = \begin{bmatrix} x[k] \\ \dot{x}[k] \end{bmatrix} \tag{10}$$

The first element represents the signal amplitude, and the second represents its rate of change. The idea is to use the current amplitude and estimated rate of change to predict the signal at the next sampling instant.

B. State Transition Matrix

The state transition matrix is

$$F = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}. \tag{11}$$

This matrix assumes that the rate of change remains approximately constant over one sampling interval. Using this assumption, the filter predicts the next amplitude from the current state.

This approximation is useful for slowly changing signals. However, a signal containing several sinusoidal components can change more rapidly, so the accuracy of this model depends on the signal being processed.

C. Measurement Matrix

The measurement provides the signal amplitude but does not directly provide its rate of change. Therefore, the measurement matrix is selected as

$$H = \begin{bmatrix} 1 & 0 \end{bmatrix}. \tag{12}$$

The measurement equation becomes

$$Z[k] = HX[k] + v[k]. \tag{13}$$

The filter uses the measured amplitude to correct its predicted state. The rate of change is estimated indirectly through the state-space model.

D. Process Noise Covariance

The process noise covariance matrix is selected as

$$Q = \begin{bmatrix} 0.001 & 0 \\ 0 & 0.001 \end{bmatrix}. \tag{14}$$

This matrix represents uncertainty in the assumed signal model. In simple terms, it allows for the possibility that the actual signal may not follow the prediction exactly.

The values chosen for Q affect how readily the filter adjusts its estimate when the measurements differ from the predicted behavior.

E. Measurement Noise Covariance

The measurement noise covariance is selected using the estimated noise power:

$$R = P_{\text{noise}}. \tag{15}$$

The value of R represents the assumed uncertainty in the measured signal. When measurement noise is expected to be large, the filter generally relies more on its prediction. When the measurement is considered more reliable, it gives the measurement greater influence.

The choice of Q and R is important because both values affect the Kalman gain and, consequently, the final estimate.

V. Kalman Filter Algorithm

The Kalman filter processes the noisy signal one sample at a time. At each step, it predicts the state, checks the difference between the prediction and the new measurement, and updates the estimate.

A. Prediction Step

The predicted state is calculated as

$$X_{k|k-1} = FX_{k-1|k-1}. \tag{16}$$

The predicted covariance is

$$P_{k|k-1} = FP_{k-1|k-1}F^T + Q. \tag{17}$$

The first equation predicts the next state using the previous estimate. The second equation estimates how uncertain that prediction is after accounting for process noise.

B. Kalman Gain

The Kalman gain is calculated as

$$K_k = P_{k|k-1}H^T \left(HP_{k|k-1}H^T + R\right)^{-1}. \tag{18}$$

The Kalman gain determines how strongly the new measurement affects the predicted state. Its value depends on the uncertainty in the prediction and the assumed uncertainty in the measurement.

C. Correction Step

The difference between the actual measurement and its predicted value is called the measurement residual:

$$r_k = Z_k - HX_{k|k-1}. \tag{19}$$

The updated state is calculated using

$$X_{k|k} = X_{k|k-1} + K_k r_k. \tag{20}$$

The covariance is updated as

$$P_{k|k} = (I - K_k H)P_{k|k-1}. \tag{21}$$

After the correction step, the first element of the updated state vector is stored as the estimated signal amplitude. The second element gives the estimated rate of change.

The same procedure is repeated for every sample until the entire noisy signal has been processed.

VI. MATLAB Implementation

MATLAB is used to generate the signal, add noise, run the Kalman filter, and examine the results. It also provides functions for calculating the performance measures and plotting the waveforms.

The simulation follows these steps:

1) Set the sampling frequency and number of samples.
2) Generate the original signal from the three sinusoidal components.
3) Calculate the noise power required for the target input SNR.
4) Add Gaussian noise to obtain the noisy signal.
5) Initialize the state vector and covariance matrices.
6) Predict and correct the state for every sample.
7) Store the estimated signal and its rate of change.
8) Calculate MSE, RMSE, and SNR for the noisy input and filtered output.
9) Plot the signals in the time and frequency domains.

One advantage of this approach is that the estimate can be updated as each measurement arrives. However, the quality of the result still depends on the accuracy of the model and the selected covariance values.

The simulation also makes it possible to compare the filter output directly with the known original signal. This comparison is useful for identifying whether the chosen configuration actually reduces the reconstruction error.

VII. Time-Domain Results

The time-domain plot provides the first indication of how the filter behaves. It shows the original signal, the noisy measurement, and the output obtained after applying the Kalman filter.

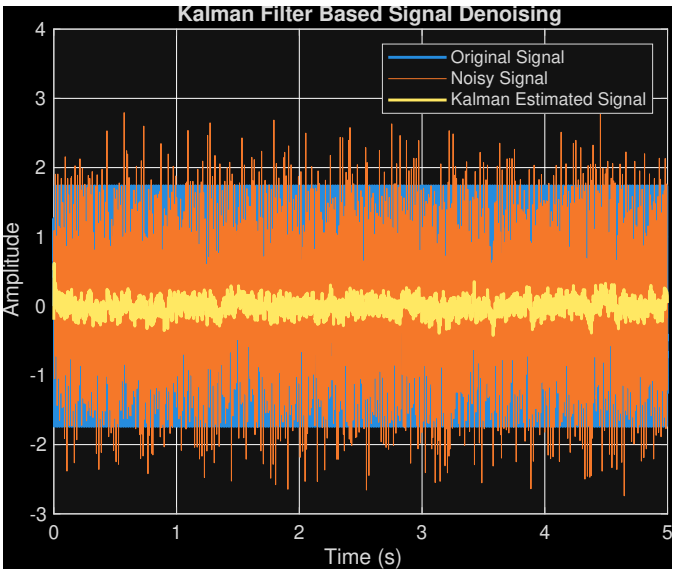


Fig. 1. Comparison of the original, noisy, and Kalman-estimated signals in the time domain.

The original waveform is used as the reference for judging the output. The noisy signal contains random variations because Gaussian noise has been added to it. The filtered waveform shows how the selected Kalman filter configuration responds to those variations.

From the comparison, the filter does not reproduce the original waveform accurately enough to provide the expected improvement. This observation agrees with the numerical results, which show that the reconstruction error increases after filtering.

A likely reason is that the two-state model is too simple to describe the behavior of the three-frequency input signal. The result suggests that the model and its parameters need to be improved before the filter can be used effectively for this particular signal.

VIII. Zoomed Time-Domain Analysis

The complete time-domain plot gives an overall view of the signal, but some differences between the waveforms can be difficult to see. To examine these differences more closely, the first 500 samples are plotted separately.

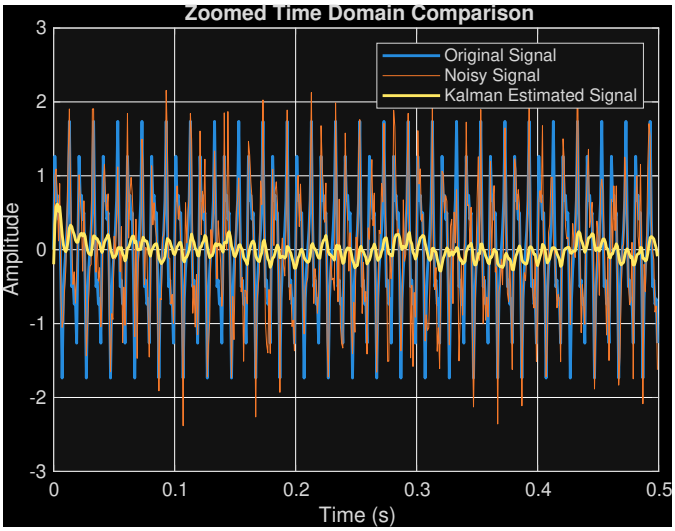


Fig. 2. Zoomed comparison of the signals over the first 500 samples.

The zoomed plot makes it easier to compare the peaks, troughs, and rapid changes in the three waveforms. It also shows whether the filtered signal follows the original signal closely or departs from it.

The input contains components at three different frequencies, so its slope changes throughout the sampling interval. The selected model assumes that the rate of change remains approximately constant over each step. This assumption may not be sufficient to represent the actual signal accurately.

Consequently, the filter can produce an estimate that differs from the original waveform instead of removing noise while preserving the useful signal components.

IX. Frequency-Domain Analysis

The time-domain plots show how the signal changes with time. Frequency-domain analysis provides another way to

study the filter by showing the frequency components present in each signal.

The Fourier transform is calculated in MATLAB using the `fft` function:

$$X(f) = \mathrm{FFT}\{x[k]\}. \tag{22}$$

Similarly, the transforms of the noisy and estimated signals are

$$Y(f) = \mathrm{FFT}\{y[k]\} \tag{23}$$

and

$$K(f) = \mathrm{FFT}\{\hat{x}[k]\}. \tag{24}$$

The magnitude spectrum can be calculated by normalizing the Fourier transform:

$$|X(f)| = \frac{|\mathrm{FFT}(x[k])|}{L} \tag{25}$$

where L represents the number of samples. A single-sided amplitude spectrum requires the usual scaling of positive frequency components.

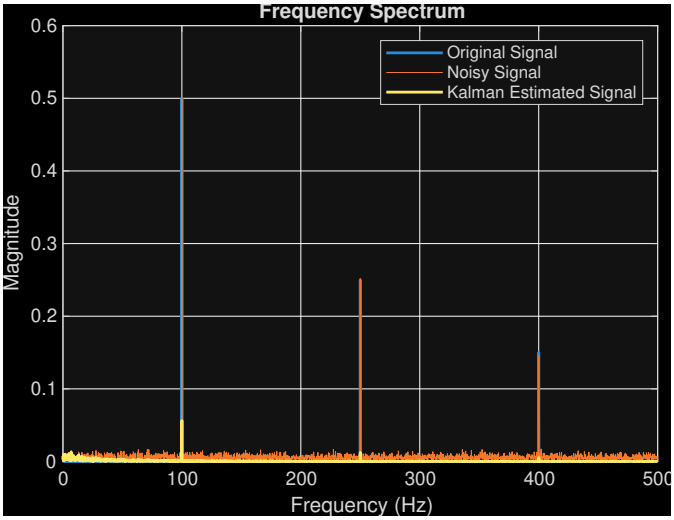


Fig. 3. Frequency spectra of the original, noisy, and Kalman-estimated signals.

The original signal contains three main frequency components at 100 Hz, 250 Hz, and 400 Hz. Once noise is added, extra spectral energy appears across the frequency range.

The estimated spectrum can be compared with the original spectrum to check whether the main components are preserved. If the filter reduces the amplitude of these components too much, the output may differ significantly from the original signal.

For this reason, frequency-domain results should not be considered on their own. They need to be interpreted along with the waveform plots and the numerical error measurements.

X. Kalman State Estimation

The Kalman filter estimates both the signal amplitude and its rate of change. Although only the amplitude is directly measured, the second state is useful because it helps the model predict how the signal may change in the next sample.

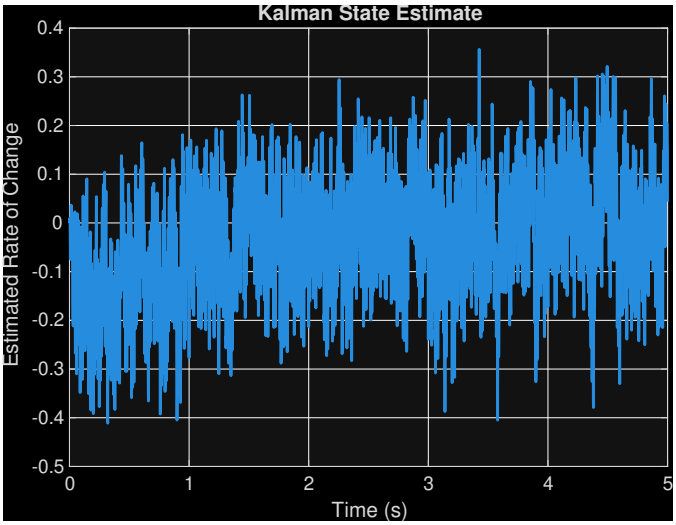


Fig. 4. Estimated rate of change obtained from the Kalman state vector.

The second state is updated at every step along with the amplitude estimate. Ideally, it should provide a useful description of how quickly the signal is changing.

However, the input signal contains several sinusoidal components, and its rate of change varies continuously. If the model cannot represent these variations accurately, the state estimate may not follow the actual signal behavior. This can also affect the amplitude estimate and contribute to the overall reconstruction error.

XI. Performance Evaluation

Visual inspection is useful, but it does not provide a complete picture of filtering performance. Numerical measures are also needed to determine whether the estimated signal is actually closer to the original.

In this project, MSE, RMSE, and SNR are used to compare the noisy input with the Kalman output.

A. Mean Square Error

Mean Square Error measures the average squared difference between the original and estimated signals:

$$MSE = \frac{1}{L}\sum_{k=1}^{L}\left(x[k] - \hat{x}[k]\right)^2. \tag{26}$$

Here, L is the number of samples and $\hat{x}[k]$ is the estimated signal.

A smaller MSE means that the estimated signal is closer to the original signal on average. A larger value indicates that the estimate contains a greater reconstruction error.

B. Root Mean Square Error

Root Mean Square Error is obtained by taking the square root of the MSE:

$$RMSE = \sqrt{MSE}. \tag{27}$$

RMSE is expressed in the same units as the signal amplitude. It provides a convenient way to understand the typical size of the estimation error.

C. Signal-to-Noise Ratio

SNR compares the power of the original signal with the power of the error:

$$SNR = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{error}}} \right). \tag{28}$$

The error power is calculated as

$$P_{\text{error}} = \frac{1}{L} \sum_{k=1}^{L} (x[k] - \hat{x}[k])^2. \tag{29}$$

For the noisy signal, the error is measured between the original and noisy waveforms. For the Kalman output, the error is measured between the original and estimated waveforms.

Using the same reference signal for both calculations allows the results to be compared fairly. If the filtering process reduces the error, the effective SNR should generally increase. If the error becomes larger, the effective SNR decreases.

XII. Quantitative Results

The numerical results obtained from the MATLAB simulation are summarized in Table II.

TABLE II
QUANTITATIVE PERFORMANCE COMPARISON

Metric	Noisy Signal	Kalman Output
MSE	0.209776	0.626822
RMSE	0.458013	0.791721
SNR	5.04 dB	0.29 dB

The results show that the noisy signal has an MSE of 0.209776 and an RMSE of 0.458013. After applying the current Kalman filter, the MSE increases to 0.626822 and the RMSE increases to 0.791721.

This means that the estimated signal differs from the original signal more than the noisy input does. In other words, the current configuration has not achieved its main objective of improving the reconstruction accuracy.

The effective SNR also decreases from 5.04 dB to 0.29 dB. The change in SNR is calculated as

$$\Delta SNR = SNR_{\text{Kalman}} - SNR_{\text{Noisy}}. \tag{30}$$

Substituting the reported values gives

$$\Delta SNR = 0.29 - 5.04 = -4.75 \text{ dB}. \tag{31}$$

The reduction of 4.75 dB indicates that the filtered output has a lower effective SNR than the noisy input. This result is an important part of the experiment because it shows that simply applying a Kalman filter is not enough to guarantee better signal quality.

The model and its parameters need to be chosen according to the characteristics of the signal being processed.

XIII. Discussion

The main observation from this experiment is that the Kalman filter does not improve the signal under the selected configuration. This may initially seem unexpected because Kalman filtering is widely used for estimation and noise reduction. However, the result becomes easier to understand when the assumptions of the model are considered.

The filter represents the signal using two states: amplitude and rate of change. This is a simple way to describe a signal whose behavior changes gradually. The test signal, however, contains three sinusoidal components at 100 Hz, 250 Hz, and 400 Hz. Their combination produces continuous changes in both amplitude and slope.

The selected state transition matrix assumes that the rate of change remains approximately constant over each sampling interval. Although this assumption simplifies the calculations, it may not represent the behavior of the combined waveform accurately enough.

The covariance matrices also affect the result. The process noise covariance is fixed at

$$Q = \begin{bmatrix} 0.001 & 0 \\ 0 & 0.001 \end{bmatrix}. \tag{32}$$

The measurement noise covariance is based on the estimated noise power. These values determine how much the filter trusts its model and how much it trusts the incoming measurement.

If the filter relies too strongly on an inaccurate prediction, it may move away from the actual signal. On the other hand, if it relies too heavily on noisy measurements, it may not remove enough noise. Finding a suitable balance is therefore an important part of designing a Kalman filter.

The numerical results confirm that the present settings need improvement. Both MSE and RMSE increase after filtering, and the effective SNR decreases by 4.75 dB.

The useful lesson from this experiment is that the quality of a Kalman filter depends on the relationship between its mathematical model, covariance settings, and input signal. A more suitable state model and carefully selected parameters should be investigated before drawing conclusions about the effectiveness of the method itself.

XIV. Limitations

The experiment provides a basic demonstration of Kalman filtering, but there are a few limitations to consider.

First, the model uses only two state variables: signal amplitude and rate of change. This makes the implementation simple, but it may not be sufficient for a signal containing multiple sinusoidal components.

Second, the process noise covariance matrix uses fixed values. These values were not optimized for the particular test signal, so they may not provide the best balance between prediction and measurement.

Third, the measurement noise covariance is based on the estimated noise power. In an actual system, the noise level may not be known in advance and can change over time.

The initial state and covariance are also selected using simple assumptions. Different initial conditions may produce different behavior, particularly during the first few samples.

Finally, the experiment uses simulated Gaussian noise. Real signals may contain interference, impulsive disturbances, or noise that does not follow a Gaussian distribution. Testing with real measurements would provide a better understanding of the filter's performance in practical applications.

XV. Future Work

The current results suggest several ways in which the implementation could be improved.

1) The process noise covariance matrix Q can be tuned using repeated simulations to identify values that provide a better estimate.
2) The measurement noise covariance R can be estimated or updated when the noise conditions change.
3) A higher-order state-space model can be developed to represent the sinusoidal components more accurately.
4) Adaptive Kalman filtering can be investigated when the signal or noise characteristics vary with time.
5) The filter can be tested using real sensor or communication signals instead of relying only on simulated Gaussian noise.
6) Its performance can be compared with moving-average, FIR, IIR, and Wiener filters using the same input signal and noise conditions.
7) The algorithm can be optimized for real-time implementation on an FPGA or embedded processor.

These changes would help determine which model and parameter settings are most suitable for the signal. They would also make the comparison more useful for applications where reliable signal estimation is required.

XVI. Conclusion

This project examined the use of a Kalman filter to reduce noise in a discrete-time signal using MATLAB. The test signal was formed by combining sinusoidal components at 100 Hz, 250 Hz, and 400 Hz. Gaussian noise was added with a target input SNR of 5 dB, and a two-state Kalman filter was used to estimate the original waveform.

The filter operated through repeated prediction and correction steps. It used the signal amplitude and its rate of change as state variables, while the process and measurement covariance matrices controlled the influence of the model and the noisy measurements.

The results were evaluated using time-domain plots, frequency spectra, MSE, RMSE, and SNR. The noisy input had an MSE of 0.209776 and an RMSE of 0.458013. After filtering, these values increased to 0.626822 and 0.791721, respectively. The effective SNR also decreased from 5.04 dB to 0.29 dB, representing a reduction of 4.75 dB.

These results show that the chosen filter configuration does not improve the reconstruction quality of the test signal. The simple state model may not describe the multi-frequency waveform accurately, and the covariance settings may not provide a suitable balance between prediction and measurement.

The experiment therefore demonstrates an important point: a Kalman filter must be designed according to the behavior of the signal it is intended to process. Choosing a suitable state-space model and adjusting the filter parameters are essential steps in obtaining a useful estimate. Further work involving model improvement, parameter tuning, and adaptive filtering could help achieve better noise-reduction performance.

References

[1] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

[2] R. G. Brown and P. Y. C. Hwang, *Introduction to Random Signals and Applied Kalman Filtering*, 4th ed. Wiley, 2012.

[3] P. S. Maybeck, *Stochastic Models, Estimation, and Control*, vol. 1. Academic Press, 1979.

[4] Y. Bar-Shalom, X. R. Li, and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation*. Wiley, 2001.

[5] S. Haykin, *Adaptive Filter Theory*, 4th ed. Prentice Hall, 2001.

[6] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Pearson, 2010.

[7] P. D. Welch, "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms," *IEEE Transactions on Audio and Electroacoustics*, vol. 15, no. 2, pp. 70–73, 1967.

[8] MathWorks, "MATLAB Documentation: Signal Processing and Filtering," The MathWorks, Inc.